

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: Computer Networks

COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure

CONCEPT MAPPING

Code of Conduct:

I V.Sheershika(192472056) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

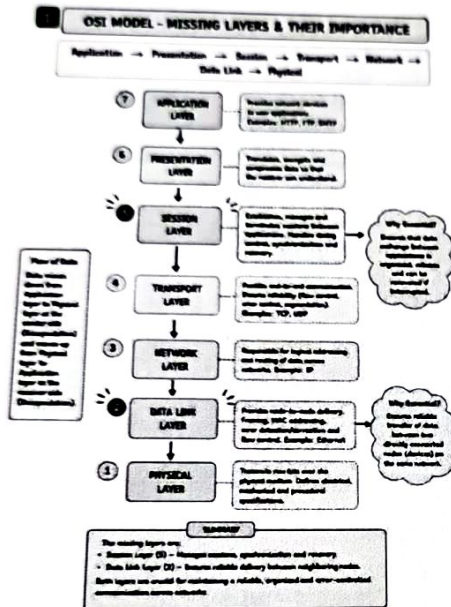
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Sheershika

COMPUTER NETWORKS – CONCEPT MAPPING

Question 1: OSI Model – Missing Layers and Their Importance

The concept map completes the OSI model by identifying the missing Session Layer and Data Link Layer. It explains the role of each layer and shows how the Session and Data Link layers contribute to reliable communication.



Concept Map Architecture

Source Concept

Application Layer

Presentation Layer

Session Layer

Transport Layer

Network Layer

Data Link Layer

Physical Layer

Relationship / Target Concept

Provides network services to users

Translates and encrypts data

Manages communication sessions

End-to-end delivery

Routing and logical addressing

Framing, MAC addressing, error detection

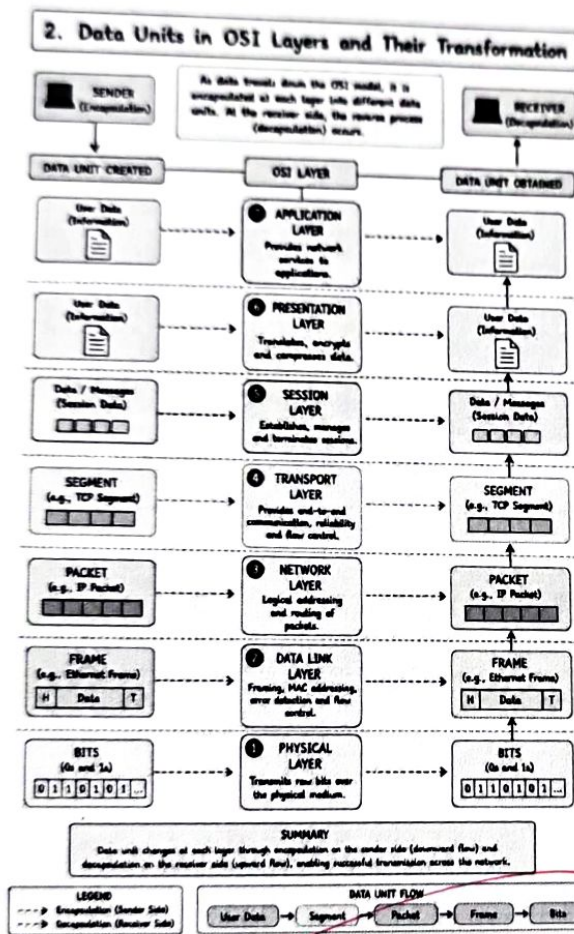
Bit transmission

Literature Insights & Evidence

- Tanenbaum & Wetherall (2021): OSI model and protocols.
- Forouzan (2022): Functions of all OSI layers.
- Kurose & Ross (2021): Reliable communication using layered architecture.

Question 2: Data Units and Their Transformation in the OSI Model

The concept map shows how data changes into Segments, Packets, Frames and Bits while passing through the OSI layers.



Concept Map Architecture

Source Concept

Transport

Network

Data Link

Physical

Relationship / Target Concept

Segment

Packet

Frame

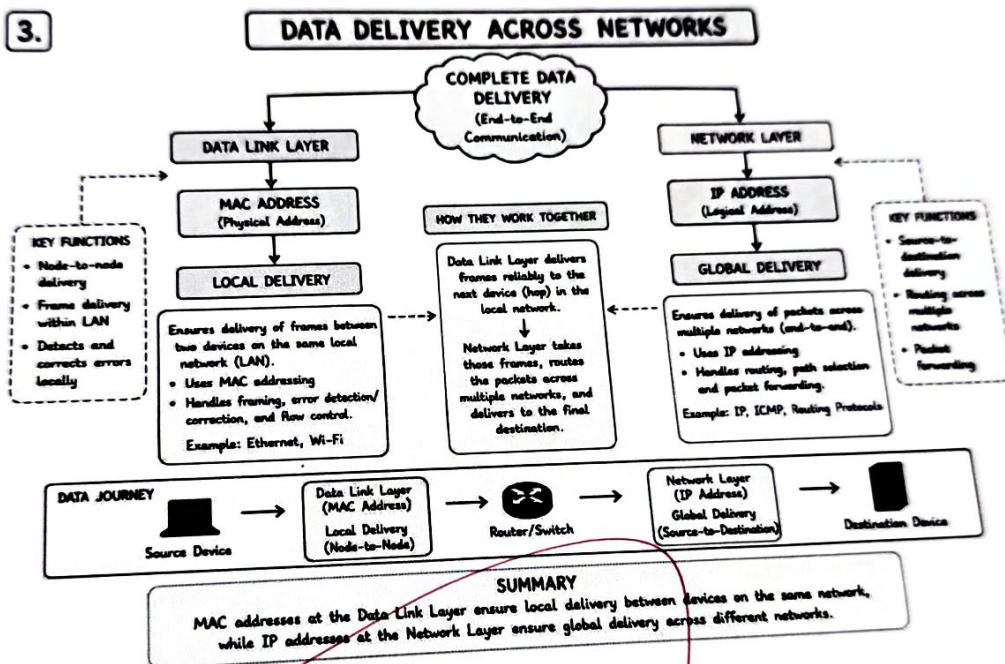
Bits

Literature Insights & Evidence

- Forouzan (2022): Encapsulation and decapsulation.
- Kurose & Ross (2021): Data units in network communication.
- Tanenbaum & Wetherall (2021): Protocol layering.

Question 3: Local and Global Data Delivery Using MAC and IP Addresses

The concept map explains how MAC addresses provide local delivery while IP addresses enable global delivery across networks.



Concept Map Architecture

Source Concept

Data Link Layer

Network Layer

Combined

Relationship / Target Concept

MAC Address → Local Delivery

IP Address → Global Delivery

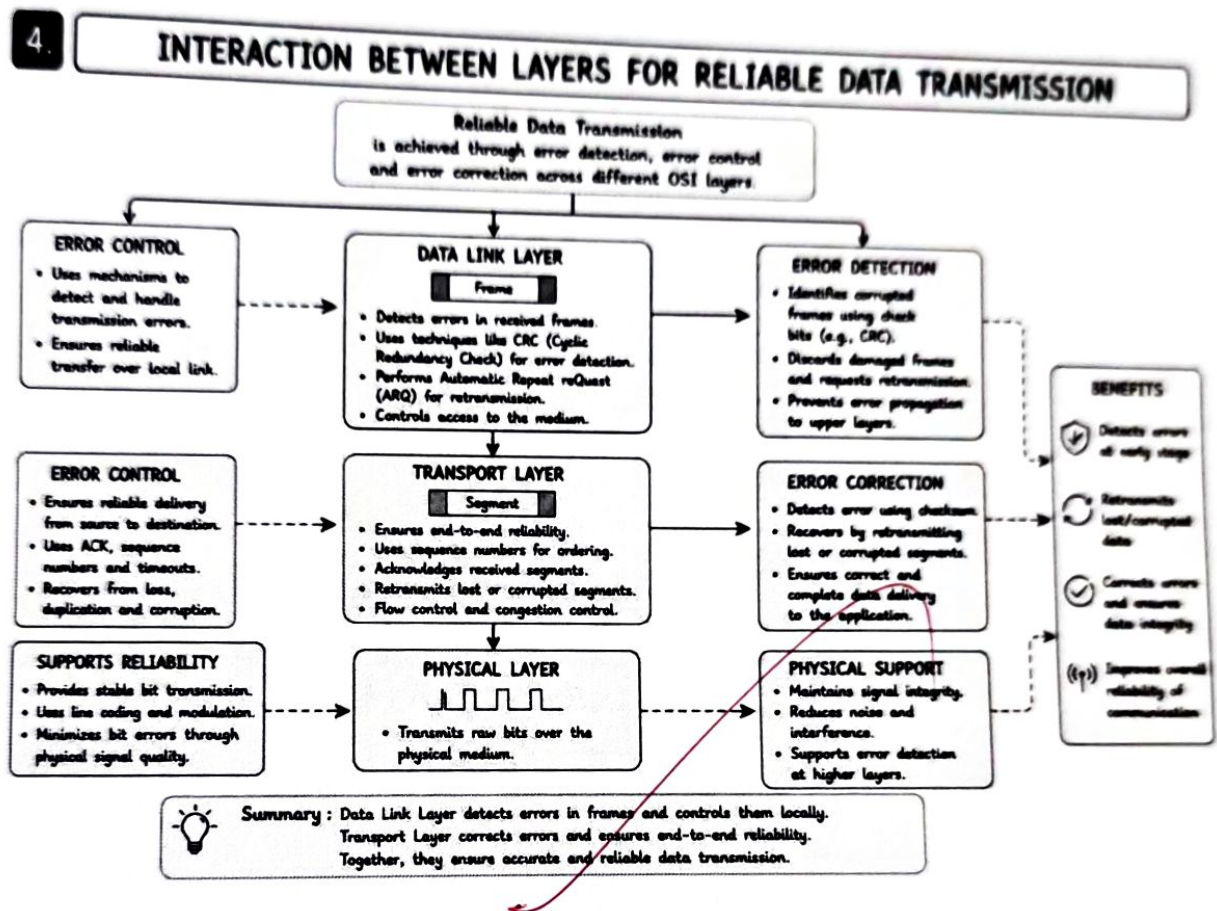
Complete end-to-end communication

Literature Insights & Evidence

- Kurose & Ross (2021).
- Forouzan (2022).
- Tanenbaum & Wetherall (2021).

Question 4: Error Control for Reliable Data Transmission

The concept map explains error detection by the Data Link layer and error correction by the Transport layer to improve reliability.



Concept Map Architecture

Source Concept

Relationship / Target Concept

Data Link Layer

Error Detection

Transport Layer

Error Correction

Result

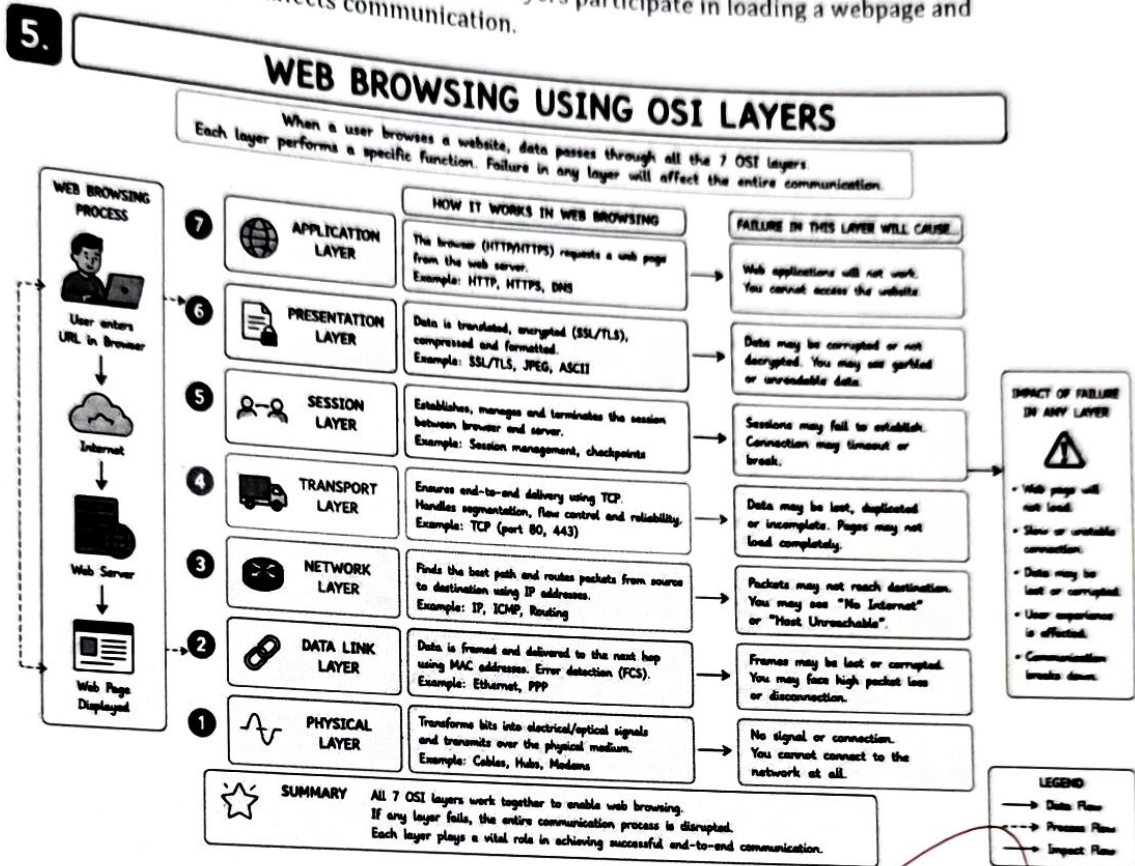
Reliable transmission

Literature Insights & Evidence

- Forouzan (2022).
- Kurose & Ross (2021).
- Tanenbaum & Wetherall (2021).

Question 5: Web Browsing Using the OSI Layers

The concept map illustrates how all seven OSI layers participate in loading a webpage and how failure in one layer affects communication.



Concept Map Architecture

Source Concept

Application

Transport

Network

Data Link

Physical

Relationship / Target Concept

HTTP Request

TCP

IP

Frame

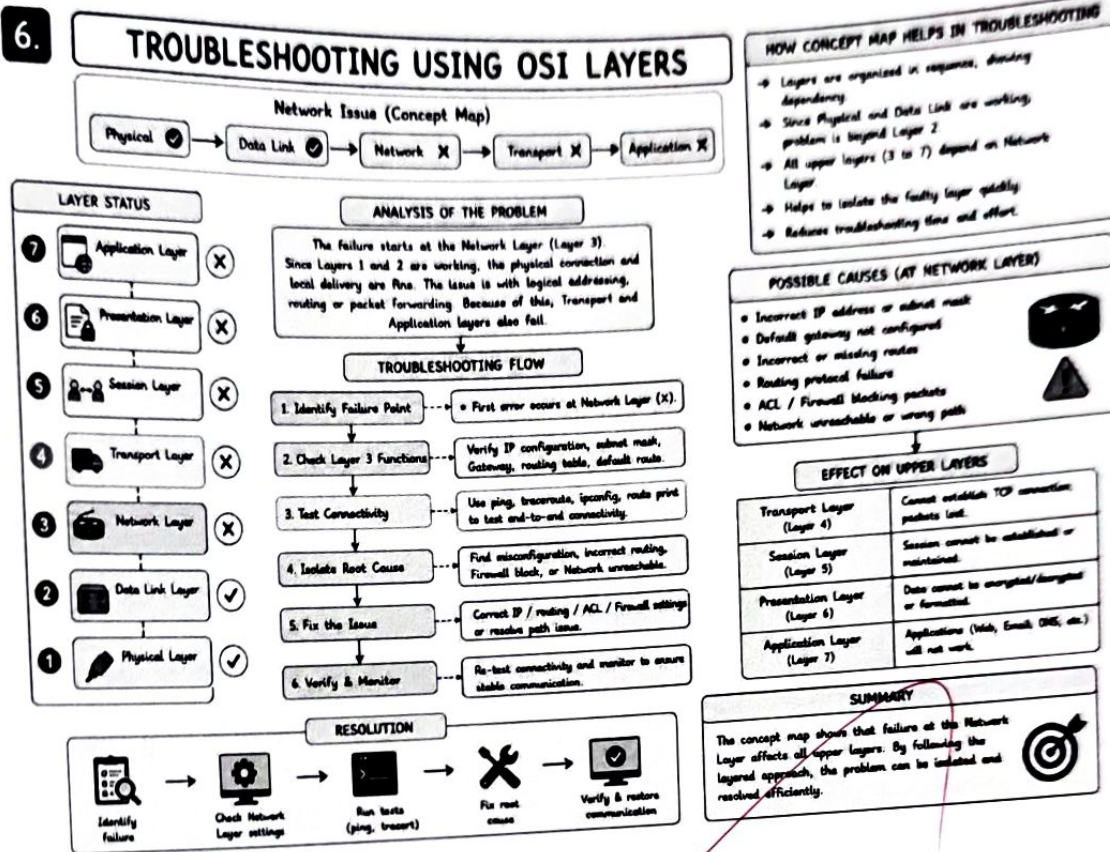
Bits

Literature Insights & Evidence

- Kurose & Ross (2021).
- Forouzan (2022).
- Tanenbaum & Wetherall (2021).

Question 6: Troubleshooting Using the OSI Model

The concept map shows a failure at the Network layer and explains how it affects upper layers, helping identify faults quickly.



Concept Map Architecture

Source Concept

Physical

Data Link

Network

Transport

Application

Relationship / Target Concept

Working

Working

Failure

Failure

Failure

Literature Insights & Evidence

- Forouzan (2022).
- Kurose & Ross (2021).
- Tanenbaum & Wetherall (2021).

Handwritten signature and date: 20/11/2023